Silence to Solidarity: How Communication about a Minority Affects Discrimination

Duncan Webb

Nova School of Business and Economics

People predominantly talk within their identity groups. Conversations between majority-group members may, therefore, be crucial in shaping discrimination against minorities. In an experiment in India (N = 3,397), nontransgender participants strongly discriminate against transgender workers when hiring for a grocery delivery, but a discussion with two nontransgender neighbors eliminates this discrimination in private postdiscussion choices. The discussion is 1.7 times more effective at reducing discrimination than information about transgender people's legal rights, and effects partially persist after 1 month. The evidence is consistent with norm-based persuasion, in which pro-transgender participants are more vocal and create a perceived anti-discriminatory norm that reduces subsequent discrimination.

I. Introduction

Discriminatory behavior harms equity and efficiency in a wide range of economic domains, including in firms (Hjort 2014; Glover, Pallais, and Pariente 2017; Hedegaard and Tyran 2018), the labor market (Charles

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and Guryan 2008; Hsieh et al. 2019; Ashraf et al. 2022; Folke and Rickne 2022), housing (Christensen and Timmins 2023), and informal social interactions (Lowe 2020). Standard theories of discrimination frame it as an individual's decision, based on beliefs or deep-seated preferences that are difficult to change (Becker 1957; Phelps 1972; Arrow 1973; Aigner and Cain 1977). But at the societal level, we sometimes observe rapid changes in discriminatory preferences (Kuran 1987; Fernández 2013; Sunstein 2019). For example, multiple countries have seen rapid increases in the proportion of people who accept interethnic marriage, homosexuality, and equal rights for women over the course of a single generation.[1] Some of these changes may be driven by social contact *between* majority and minority groups (Allport 1954; Paluck, Green, and Green 2019). But in a world of homophily, where people mostly interact with others within their identity group (Jackson 2011), communication *within* the majority group may also be a powerful determinant of change.

In this paper, I therefore test how communication between majority-group members about a minority (*horizontal communication*) affects discrimination. Standard models typically predict that communication merely averages out individual beliefs, leading beliefs to become more similar within a group but not changing the mean belief or resulting average behavior.[2]

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[1] For example, (i) the proportion of people in the United States saying that homosexuality is wrong dropped rapidly in a single year (1993) and continued to decline quickly in the 30 following years (Fernández, Parsa, and Viarengo 2019); (ii) the proportion of people in the UK indicating discomfort with interethnic marriage dropped from 55% in 1983 to 25% in 2013 (Park, Bryson, and Curtice 2014); and (iii) the proportion of people in Uganda saying that women should have equal rights went from 63% in 2002 to 80% in 2012 (Chingwete, Richmond, and Alpin 2014).

[2] For example, agents who truthfully share information will eventually converge to the same posterior belief in standard Bayesian models (Aumann 1976; Geanakoplos and Polemarchakis 1982). And if agents' priors are not systematically incorrect, the martingale property of Bayesian updating implies that their beliefs should not systematically shift in one direction.

Yet horizontal communication could lead to large shifts in postcommunication discrimination if there are asymmetries between pro- and antiminority communication. For example, pro-minority individuals might be more persuasive, using moral arguments in their favor; antiminority people might have stronger preferences against minorities and so choose to speak up more in discussions; or antiminority individuals may be unwilling to express discriminatory views if they are concerned about how they are perceived by others. Such asymmetries could drive changes in postcommunication discrimination if communication affects people's attitudes, beliefs, or the norm of behavior toward a minority.

I run a field experiment in urban Chennai, India ($N = 3{,}397$), that tests these ideas in the context of discrimination against the most visible LGBTQ+ group in India: a community of transgender women known locally as *thirunangai*. This setting is an appropriate context in which to study the effect of communication on discrimination. The community is vulnerable to extensive economic discrimination and violence (US State Department 2021), and the *thirunangai*'s distinct visual identity and historic role in Indian society make them highly recognizable—allowing me to measure discrimination by showing participants photos. At the same time, there appears to be nascent social change toward greater acceptance of transgender people. This may create conditions in which communication reduces discrimination; for example, despite strong de facto discrimination, there is also widespread agreement that discrimination is wrong.[3] This raises the possibility that pro-trans individuals may be more vocal or more persuasive in such settings.

I first evaluate whether horizontal communication can affect discrimination by randomizing whether participants are involved in a group discussion with two of their neighbors. I measure the effect of this discussion on anti-transgender discrimination in a series of private, individual hiring choices after the discussion has ended. Participants are offered a free grocery delivery and make a series of binary choices over the worker who will carry out the delivery (along with the items they will receive, which are randomly varied across choices). Participants who do not take part in a discussion are highly discriminatory: they are 19 percentage points (32%) less likely to hire transgender workers than nontransgender workers ($p < .001$). Their choices imply that they are willing to sacrifice grocery items worth 1.9 times the median daily food expenditure to avoid interacting with a transgender worker for 15 minutes.

[3] Despite observing substantial hiring discrimination in my control group, 93% of that same control group say that discrimination is unacceptable in response to a vignette that showcases explicit discrimination. There appears, therefore, to be a wedge between the descriptive norm (how much people actually discriminate) and the prescriptive norm (to what extent people think it is right or wrong to discriminate).

Horizontal communication leads to large short-run reductions in subsequent discrimination: in private choices after a discussion, there is no discrimination against transgender workers on average. In the discussion condition, participants discuss a series of hiring options as a group of three neighbors and are asked to make collective hiring choices.[4] Since some of these options include transgender workers, participants naturally discuss whether to hire transgender workers. Crucially, the only communication about transgender people in this discussion comes from the participants themselves, rather than from the discussion facilitator.

The effects of this discussion on discrimination are stark: in people's private, postdiscussion hiring choices, participants are 17 percentage points (42%) more likely to select a transgender worker than the control group ($p < .001$), implying that anti-transgender discrimination is reduced to 0 on average (p of difference between transgender and nontransgender: .30). The effects are also partially persistent, although there is substantial fade-out: when I resurvey participants approximately 1 month later, discussion participants are 4 percentage points more likely to select transgender workers than the control group in a series of hypothetical hiring choices ($p = .03$).

I then compare the effects of the discussion with communication about a minority from an agent of authority, namely, the legal system (*top-down communication*). I cross-randomize whether participants watch a video informing them about an Indian Supreme Court ruling that affirmed that transgender people have all the same fundamental rights as other citizens, including freedom from discrimination.[5] Such communication could affect discrimination if it signals that discrimination is not socially acceptable (Sunstein 1996; McAdams and Rasmusen 2004; Bénabou and Tirole 2011) or if participants fear legal enforcement. Learning about these legal rights also lowers discrimination by 10.3 percentage points ($p < .001$). But the effects are only 59% as large as the effects of the group discussion (p of difference $< .05$) and do not persist after 1 month. In this context, therefore, horizontal communication about transgender people is substantially more effective than top-down information about the law.

[4] Including hiring choices in the discussion reduces the possibility of cheap talk by increasing the stakes of the discussion. It also implies that I measure the bundled effect of (i) observing others' choices and (ii) listening to the arguments made in the discussion. I later show evidence that both matter: observing others' choices without a discussion moderately reduces discrimination, while features like how much *morality* was invoked in the arguments correlate with reduced discrimination even when controlling for choices.

[5] Groups were always same-gender to avoid overly hierarchical relationships between group members. To test for the effects of *within-network* horizontal communication, to make the discussions more naturalistic, and to ensure social image concerns could affect behavior, we recruited neighbors who knew each other 98% of the time.

There are no interaction effects between the legal rights video and the group discussion ($p \in [.96, 1.00]$). The effect of the discussion is also not driven by interaction effects; it is robust to only using the sample who were not informed about transgender rights.

I then seek to understand why discussions reduce discrimination in this context. The suggestive evidence is consistent with *norm-based persuasion* as the primary mechanism, in which pro-transgender participants speak up more during discussions, use especially persuasive moral arguments, and thus generate a perception of a strong antidiscriminatory norm that persuades others to discriminate less in their subsequent private choices.[6]

I first use a mechanism treatment to show the effects are driven by interpersonal persuasion rather than self-persuasion (Schwardmann, Tripodi, and van der Weele 2022). Silent "listeners," who listen to but do not participate in discussions, are 12.5 percentage points more likely to select a transgender worker afterward ($p < .001$)—an effect statistically indistinguishable from that of active participants. This finding supports interpersonal persuasion since listeners do not speak in the discussion and yet still show treatment effects as strong as participants. Consistent with this, the more pro-transgender statements a listener hears from others, the less they subsequently discriminate.

Second, I use correlational evidence to show that persuasion appears to operate through changes in perceived descriptive norms—shifts in what people expect others to do. Norm updates alone suffice to reduce discrimination: in a treatment without discussions, "observers" learn their group members' first-round choices, shift their beliefs about within-group norms, and subsequently discriminate less. Specifically, participants predict that their groups will privately discriminate 4.6 percentage points less if they have observed others' choices. This norm update is accompanied by an estimated 4.2 percentage point reduction in discrimination in observers' own private second-round choices. This direct relationship between perceived norms and postdiscussion discrimination confirms that updating perceived descriptive norms—even without the rich interaction of discussion—can reduce discrimination. I then show that the discussion generates large shifts in these perceived norms (24 percentage points, $p < .001$), while producing minimal changes in attitudes or beliefs about transgender workers. Since these norm shifts strongly mediate the discussions' effects on discrimination, this is consistent with norm updates being a key driver of the discussion's effects.

Norm-based persuasion can explain how participants influence each other's private behavior but not why discussions create a large mean shift *away* from discrimination in this context. Standard communication models predict that communication should "average out" behavior; that is, behavior should converge within a group, with no systematic change in mean behavior (see sec. VI). The substantial reduction in discrimination therefore requires sources of *asymmetry*, whereby pro-transgender and

[6] The implied *persuasion rate* (29%) falls at the upper end of the range in the literature (DellaVigna and Gentzkow 2010) but remains consistent with other successful in-person interventions and is plausible given the short-term nature of the effects.

anti-transgender forces do not simply cancel out. I use correlational evidence to document three potential sources of such asymmetry.

First, privately pro-trans participants speak up more frequently during discussions. Those who privately select more transgender workers after discussions (a proxy for pro-trans private preferences)[7] are 32% more likely to speak first and 27% more likely to dominate in discussions involving transgender workers. This asymmetric behavior appears to be motivated by a particular desire to influence others, rather than by social image concerns—pro-trans participants are more likely to self-report intentions to shape others' choices than other participants. Pro-trans statements thus occur 2.5 times more frequently than anti-trans statements in the transcripts. This is also driven by the substantial minority of pro-trans participants: despite strong average discrimination, 31% of control participants still have a positive willingness to pay to select a transgender worker.

Second, I show suggestive evidence of asymmetric inference about norms: participants more readily shift toward a perception of pro-trans norms than away from them. Discussions *generate* misperceptions—participants overestimate how pro-trans their group members are after a discussion, reversing the underestimation seen in the control group. This suggests that participants fail to account for pro-trans participants' greater propensity to speak up, thus overestimating the prevalence of pro-trans behavior. Descriptive evidence also suggests that participants exhibit *silence neglect*, updating beliefs when hearing explicit pro-trans statements but neglecting the information contained in silence. Participants who hear *no* pro-transgender statements during discussions show no change in perceived norms or behavior, suggesting that silence carries little informational weight, even though the control group's results show that silent participants are largely anti-trans. These patterns are consistent with a "spiral of silence" (Noelle-Neumann 1974; Ho and Huang 2026), in which selective communication leads people to perceive a view as more prevalent than it truly is. They also link to work on misperceived social norms (Bursztyn, Egorov, and Fiorin 2020), but with a distinctive feature—communication itself generates the misperception.

Finally, I analyze the discussion transcripts to unpack the black box of the discussion and show suggestive evidence of *rhetorical asymmetry* (Schkade, Sunstein, and Kahneman 2002), in which some arguments are more persuasive than others. In particular, the evidence is consistent with

[7] I did not include baseline measures of discrimination in order to minimize any priming or experimenter demand effects before eliciting the main discrimination outcome, so only postdiscussion choices are used in this analysis. Baseline and endline attitudes are likely to be positively correlated under a reasonable assumption: that pro-trans participants are not *so* persuasive that they persuade others to be even more pro-trans than themselves. Nevertheless, the lack of baseline measures is an important limitation to this analysis.

pro-trans arguments being more persuasive than anti-trans arguments because they more often invoke moral language and principles. Participants frequently invoke moral concepts when discussing transgender workers, using language about "opportunities," "rights," and "equality." These moral arguments are correlated with both larger changes in perceived norms and greater reductions in discrimination, even after controlling for within-discussion choices.[8] These findings align with research on moral reframing, which shows that arguments framed in moral terms are especially persuasive when they resonate with listeners' values (Voelkel and Feinberg 2018; Feinberg and Willer 2019; Kalla, Levine, and Broockman 2022).

Why does horizontal communication reduce discrimination here, even though it can clearly promote it in other settings (e.g., Myers and Bishop 1970; Satyanath, Voigtländer, and Voth 2017)? While I cannot definitively identify which contextual features are essential in the absence of cross-context comparisons, the mechanism results suggest one crucial feature: the stark disconnect between the descriptive norm (widespread discriminatory behavior) and the prescriptive norm (widespread agreement that discrimination is wrong). Control group participants discriminated substantially, yet 93% described discrimination as wrong in response to vignettes. This disconnect can plausibly explain the three sources of asymmetry that could explain the shift away from discrimination: (i) pro-trans participants speaking up more because there is minimal social harm from advocating for transgender workers and more to gain from persuading others; (ii) participants more readily updating their perception of the descriptive norm in favor of pro-trans behavior to align more closely with the prescriptive norm; and (iii) participants being particularly responsive to morally framed arguments in favor of transgender workers because they align with their existing perceptions of what is "right." These suggestive patterns, though not conclusive, point to promising contexts for horizontal communication interventions where similar norm disconnects exist. They simultaneously indicate that horizontal communication is not a silver bullet and may be actively harmful where people subscribe to the view that discrimination is morally correct.

The key contribution of the paper is to show that discrimination can be rapidly reduced by generating horizontal communication about a minority within an existing social network. While many previous studies have shown that social contact between in-groups and out-groups can affect discrimination (Allport 1954; Pettigrew 1998; Boisjoly et al. 2006; Pettigrew and Tropp 2006; Paluck, Green, and Green 2019; Rao 2019; Lowe

[8] Since being in a group with others with a persuasive personality does not predict greater reductions in discrimination, this indicates that it is the *arguments* rather than the *people* that are more persuasive.

2020, 2025; Corno, La Ferrara, and Burns 2022; Bursztyn et al. 2024; Ghosh 2025), I show that even communication between in-group members about an out-group can reduce discrimination.

The most closely related studies are Broockman and Kalla (2016) and Kalla and Broockman (2020, 2023), which show that trained door-to-door and phone canvassers in the United States can reduce prejudice through short conversations, even when the canvasser is not from the minority group. My study differs in three main ways. I focus on communication *within an existing social network*—among neighbors who know each other well, rather than canvassers talking to strangers. Understanding whether existing social networks can generate norm change raises new conceptual issues: network members may exert stronger influence on each other, but also face higher costs for challenging discriminatory norms or be too homogeneous to generate changes in behavior. Second, I examine *undirected* communication, where participants have no explicit incentive or instruction to reduce discrimination, unlike trained canvassers with clear directives to decrease bias. The endogenous choice to speak up against discrimination appears to be important in my context and reveals how norms might shift organically without external intervention in other contexts. Third, I examine communication that includes making revealed-preference choices, with revealed-preference discrimination outcomes. This reduces the possibility of cheap talk both during the intervention and when measuring the outcome.

Other papers evaluate the effect of in-group contact on out-group prejudice. Scacco and Warren (2018) find that in-group contact backfires, and other papers typically find null results (see Lowe 2025). I contribute to this scarce literature by documenting a large reduction in discrimination that contrasts with previous studies and by focusing on semistructured communication about a minority among the in-group, rather than unstructured in-group contact that is often bundled with other treatments (e.g., skill training).

The discussion generates a misperceived norm that reduces discrimination, in line with work on pluralistic ignorance (misperceptions about norms) and how they are affected by communication (Kuran 1987, 1991, 1997; Bursztyn, Egorov, and Fiorin 2020; Bursztyn, González, and Yanagizawa-Drott 2020; Matavelli 2023; Ho and Huang 2026). My work differs from previous papers in three ways. First, participants generate the norm shift themselves through horizontal communication, rather than in reaction to statistics about norms delivered top-down by the experimenter. My results imply that more naturalistic communication could rapidly shift behavior, even without external intervention. Second, I expand the literature on persuasion by documenting the importance of moral arguments and the endogenous choice to persuade (DellaVigna and Gentzkow 2010; Bénabou, Falk, and Tirole 2020), highlighting the

potential role of moral "activists" in driving social change. Third, while other work implies that social change occurs only in the presence of pluralistic ignorance, my results suggest a different precondition: when there is a wedge between what people do and what they believe is right. If people discriminate even when they think it is wrong, horizontal communication is likely to be especially helpful because pro-minority communicators will be more willing to speak up, more able to credibly signal an anti-discriminatory norm, and more able to use persuasive moral arguments.

Finally, I examine potential policy levers for reducing discrimination against LGBTQ+ people in a low- or middle-income country (LMIC). Even though such discrimination may have significant costs (Badgett 2014; Badgett, Waaldijk, and Rodgers 2019), very little research in LMICs has examined its effects and causes (Badgett, Carpenter, and Sansone 2021).[9] I contribute by evaluating the potential to reduce this discrimination using horizontal communication.

II. Context: Transgender Community in India

This study examines discrimination against a historically marginalized community in South Asia largely composed of transgender women, who in the state of Tamil Nadu are called *thirunangai*.[10] This group has a long-standing cultural and religious role in Indian society (Reddy 2005; Kalra 2012). Their visually recognizable identity, however, leaves them particularly susceptible to discrimination (Sharma 2014; Agoramoorthy and Hsu 2015). Economic discrimination against this group is multifaceted. Transgender people are often excluded from traditional forms of paid employment, pushing many into poverty and sex work (Masih, Singh, and Mishra 2012; Badgett 2014; Shivakumar and Yadiyurshetty 2014; Nuttbrock 2018), resulting in low levels of formal employment (Kerala Development Society 2017). There are likely at least 0.5–1.25 million transgender people in India (Jaffrey 1996; Registrar General and Census Commissioner, India 2011; Dixit et al. 2023), implying that such discrimination results in large welfare and efficiency costs (Badgett 2014).

[9] There are notable exceptions. Lyon (2023) shows that explaining to Ugandan citizens that homosexuality is legal in other countries leads to a backlash effect, worsening participants' opinions of those countries. Gulesci, Lombardi, and Ramos (2023) show that Latin American soap operas with LGBTQ+ characters also generate backlash, and Abbate et al. (2023) show evidence of housing discrimination for some LGBTQ+ people. Research on anti-LGBTQ+ discrimination in Europe and the United States has examined its magnitude and nature (Tilcsik 2011; Flores 2015; Klawitter 2015; Burn 2020; Button et al. 2020; Carpenter, Eppink, and Gonzales 2020; Granberg, Andersson, and Ahmed 2020; Drydakis 2022; Badgett et al. 2024), along with strategies for reducing it (Tankard and Paluck 2017; Ofosu et al. 2019; Sansone 2019; Aksoy et al. 2020, 2022).

[10] Throughout the paper, for simplicity, I refer to people from this community as "transgender people."

Widespread prejudice and discrimination appear to be increasingly at odds with social norms that penalize discriminators or promote pro-trans activism. Discrimination, though common, can generate social disapproval: in the study control group, discriminatory scenarios were rated as "wrong" 93% of the time. This aligns with other survey evidence that indicates widespread support for protecting transgender people from discrimination (IPSOS 2018). The context is thus analogous to other settings where private prejudice is relatively common, but its expression may be inhibited by social sanctions for prejudiced behavior (Bursztyn, Egorov, and Fiorin 2020; Bursztyn et al. 2023).

Legal changes may have contributed to the decreasing acceptability of discrimination. In 2014, the Supreme Court recognized all constitutional rights for transgender people, along with their right to identify as a third gender, and encouraged government initiatives to combat anti-transgender stigma. But awareness of these recent changes remains low. Thirty-six percent of control participants either believe that trans individuals do not have any legal status in India or cannot identify a single legal right they hold.

The study took place in Chennai, the largest city in the state of Tamil Nadu. The urban setting is advantageous because (i) approximately 80% of *thirunangai* live in urban areas (Subramanian et al. 2015), so urban residents are aware of and can visually recognize transgender people; and (ii) urban residents are familiar with online delivery services, allowing me to use delivery service market research for the study.[11]

III. Experimental Design

A. *Design Overview*

The field experiment engaged 3,397 participants in Chennai, India, to measure the effect of horizontal communication (group discussions) on subsequent hiring discrimination against transgender workers.

The main goals of the experimental design were to (i) measure discrimination in choices with real stakes; (ii) generate horizontal communication *between* participants about transgender people, without the purpose of the study being obvious; (iii) understand the mechanisms driving the effects; and (iv) benchmark effects against the effect of delivering information about transgender people's legal rights.

[11] Eighty percent of the sample say that they have previously ordered goods to be delivered to their home using an app, reflecting the popularity of meal delivery services such as Swiggy and Zomato. The market research framing is also not so unusual in this context: 29% of the sample have previously taken part in a market research survey or received a free item as a promotion.

All treatments and primary data collection occurred in one session that lasted approximately 1 hour. To allow for a group discussion, three enumerators recruited and interviewed three respondents (a "group") at the same time.

To measure hiring discrimination, participants were offered a free grocery delivery to their home and made 10 binary choices between different delivery workers and grocery bundles. One choice was randomly selected to be implemented: between 2 and 9 weeks later, the selected worker delivered the selected grocery bundle, and participants were asked follow-up questions.

The first four choices served as treatment variation, while the final six measured outcomes. During the *treatment round*, some participants discussed their preferences in groups, while others chose privately. Then during the *outcome round*, all participants made their final six choices individually and privately. These private choices are the main outcome, allowing me to measure how discussions affected *private* behavior after the discussion.

The main treatments examined how horizontal communication affects discrimination and why. The main effect of interest compares (i) *three-person discussion*, in which all three participants had a discussion about their preferred hiring options and made collective choices; and (ii) *no discussion (private)*, a control condition in which all participants made private individual hiring choices even during the treatment round. Two further treatments explore the mechanisms: (iii) *two-person discussion*, in which two participants had a discussion and made a collective choice, while the third participant silently listened; and (iv) *no discussion (public)*, in which all participants made individual hiring choices that they knew would later be revealed to the other group members.

A secondary source of treatment variation tested the effects of top-down communication about legal rights. I cross-randomized a video shown to participants before they made any hiring choices. Participants either saw (i) a *legal rights* video containing information about the legal rights of transgender people, (ii) a *rights messaging* video containing persuasive messaging in favor of transgender rights, or (iii) a *control* video that did not mention transgender rights.

Figure 1 shows a summary of the experimental design for the main session (for further details, see fig. A1; figs. A1–L12 are available online). I describe the design further below, with additional details in appendix B (apps. A–L are available online).

B. Sample and Recruitment

Participants were recruited from urban areas in Chennai between March and July 2023 through direct household canvassing and introductions

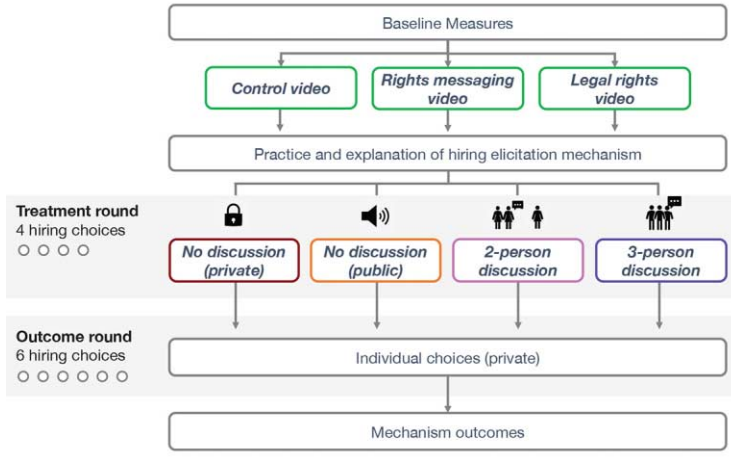


FIG. 1.—Summary of experimental design. More detail is given in figure A1.

from community leaders (see fig. A2 for survey locations). All participants were offered a free grocery delivery as an incentive to participate. Participants were aged 20–65 and could read Tamil, and the median per capita food expenditure (Rs. 2,000 per month; US$95.24 PPP) was similar to a representative 2012 sample of urban Tamil Nadu residents (NSSO 2012).

We framed the study as market research about grocery delivery preferences to minimize experimenter demand effects. Participants were truthfully told that we were trying to understand people's preferences for grocery delivery options. Only 8% of participants guessed the study's focus on transgender workers after completing their main hiring choices. These participants do not drive the treatment effects (see app. L).

To allow for the group discussion, three enumerators always recruited and interviewed three respondents at the same time (including for the control group). To avoid recruitment strategies that differed across treatments, enumerators were blind to treatment status before starting the survey.

Groups consisted of neighbors or acquaintances from the same street or locality to facilitate natural interactions. Group members knew each other 98% of the time and described each other as family or friends 41% of the time and as neighbors 62% of the time. To avoid hierarchical relationships in which one group member dominated a discussion, we recruited either all-male or all-female groups without multiple members of the same household. The sample was predominantly female (85%). The study's focus on deliveries was more relevant for females, since they were more likely to be responsible for managing food expenditures and receiving deliveries (88%) than men (59%).

C. *Hiring Choices*

Participants made 10 binary choices between delivery options, with one randomly selected to be implemented. Each option showed the workers' photos and the grocery items offered (see fig. 2), sometimes inducing a trade-off between a preferred worker and preferred items.

In each choice, there was a "benchmark" option (cisgender male) and an "alternative" option (either cisgender male, cisgender female, or transgender).[12] Throughout the paper, I measure anti-transgender discrimination as the reduction in probability of choosing the alternative when the alternative was transgender.

Four of the 20 workers were transgender. This proportion was chosen to ensure sufficient power without making the experiment's purpose too obvious. We randomized the alternative worker's position (left or right), the order of choices, and the specific photos within gender categories. We never showed the same worker twice to a participant.[13]

Participants could visually identify the transgender workers in the photos. A validation study in late 2022 ($N = 114$) showed that participants correctly identified transgender photos as being transgender 97% of the time (table A3; tables A3–L16 are available online).

I measure participants' willingness to pay to avoid certain workers by randomly varying the items offered in each choice, so in 40% of cases, one worker offered more items than the other. Each worker offered either one item (masala spice mix), two items (masala and tea), or three items (masala, tea, and ghee). The randomization was balanced across genders. The clear ranking of the bundles made the trade-off between item value and worker characteristics clear for participants. The value of the item bundles was substantial relative to participants' consumption, corresponding to 103%, 234%, and 365% of median daily per capita food expenditure, respectively.

To ensure participants anticipated some social contact with the worker, participants learned they would have a 15-minute conversation to discuss their satisfaction with the service with their chosen worker during the delivery. When selecting workers, participants were told to consider the worker's characteristics, the items they offered, and this 15-minute conversation.

[12] This reduces the number of gender combinations, thus increasing power on the male-to-trans comparison, although it does not allow a direct comparison of trans and female workers.

[13] Photos were selected from a pool of 20 males, 21 females, and 13 transgender people. Cisgender photos were of survey enumerators who carried out the main survey. Participants were never shown pictures of the enumerator team that interviewed them to avoid response bias. Transgender photos showed real enumerators we recruited who had agreed to carry out deliveries if selected but did not carry out the main survey. Later mechanism outcomes (that did not require workers to deliver goods) used stock headshot photos.



FIG. 2.—Example of one of the binary choices presented to participants.

To test whether discrimination was statistical (e.g., stemmed from beliefs about worker reliability), some choices included truthful extra signals of worker quality. These included a "reliability score" (the proportion of successful deliveries from timed training exercises), work experience, and languages spoken (see app. B). These were sampled to be balanced across genders.

Implementing choices.—To ensure incentive compatibility, enumerators used scratch cards to randomly select one of the participants' 10 choices to be implemented. The participant received the chosen items from the chosen worker 2–9 weeks later, and the worker carried out a 15-minute follow-up survey at the same time. To minimize risk to transgender workers, the randomization selected choices involving transgender workers in fewer than 1% of cases.[14]

We helped participants understand the randomization scheme using a practice round. Mimicking the main hiring choices, participants made four binary choices between items worth under Rs. 5, with enumerators using scratch cards to implement one choice. We asked comprehension checks before the practice and main hiring round. The high success rates (92% correct the first time for the practice round, 86% for the main round) indicated good understanding.

[14] This design protected transgender workers from potential abuse or violence while maintaining incentive compatibility: participants could receive a delivery from any chosen worker, but the probability was lower for transgender workers. When transgender workers did make deliveries, they were accompanied by a team of 2–3 enumerators and a supervisor, and minimized participant interaction. See app. D for discussion of the ethical considerations.

D. Treatments

1. Discussion Arms

The first four hiring choices (the *treatment round*) varied how participants communicated about their decisions. Two of these four choices included a transgender worker. Groups were randomly assigned to one of four conditions, with randomization stratified by participant gender and survey team:

1. Three-person discussion; $N = 890$. Three respondents discussed their delivery and worker preferences and then made *joint* choices (see more detail in sec. III.D.2).
2. Two-person discussion; $N = 549$. Two randomly selected "speakers" discussed and chose together, while the third (the "listener") observed silently and listened to the speakers' choices and justifications. This mechanism arm tested whether the effects were driven by interpersonal influence or whether active participation was necessary.
3. No discussion (public); $N = 599$. Participants made silent individual choices, knowing that their choices would be later announced to others in their group. This mechanism arm exogenously increases social image concerns and evaluates whether this alone affected participants' choices. I also varied the timing of the announcement to test whether observing others selecting a transgender worker could affect subsequent discrimination and how this compared with listening to a discussion. Two randomly selected "observers" learned others' choices before making their private outcome-round decisions, while the third ("nonobserver") learned others' choices only afterward.[15] These observers were designed to test whether changes in the perceived group norm of discrimination were sufficient to affect discrimination.
4. No discussion (private); $N = 1{,}359$. Participants made choices individually and privately, serving as the control condition.

Enumerator observations indicate that participants correctly followed the protocols.

2. Design of the Discussion

Treatment round activities for the three group conditions occurred with all participants together, usually in someone's home or a nearby common

[15] Participants were not told about the distinction between *observers* and *nonobservers* until after the end of the treatment round, in order to avoid this affecting treatment round choices.

area like a building courtyard. These group sessions lasted about 10 minutes. In contrast, *no discussion (private)* participants completed their interviews individually and out of earshot from one another.

Both discussion treatments required participants to make collective decisions for each pair. When these collective choices were selected by the scratch cards, each group member separately received the same items from the same worker. The collective choices created an incentive to actively participate in the discussion by ensuring the discussion had real stakes for participants. In contrast, participants in *no discussion* treatments made individual choices.

Participants freely discussed their preferences and worked to resolve disagreements about options before coming to a conclusion about which option to select. I document the types of statements made during the discussion in section VI.C. The enumerator leading the discussion never mentioned the word *transgender* themselves, to minimize demand effects and to measure only the effects of horizontal communication arising between participants (see app. C for the discussion script and further details).

3. Rights Videos

I cross-randomized a second set of treatments to compare with the effect of top-down communication about minority rights. Participants viewed one of three 80-to-90-second videos about rights before making their hiring choices. The video was narrated in Tamil by a local research team member (who was not shown). While most content remained constant across videos and explained consumer and worker rights for delivery services, the videos differed in one key example used to illustrate rights (app. E contains full scripts and further details):

1. Legal rights video; $N = 1{,}133$. This informed participants that transgender people have legally protected rights in India: "The Supreme Court of India, the most powerful legal institution in the country, gave transgender people all the same fundamental rights as others under the Constitution of India. The law therefore gives them the right to housing, employment, and education without discrimination. All these rights that you have, they also have according to the law."
2. Rights messaging video; $N = 1{,}132$. This argued that transgender people *should* have rights but did not mention legal protections, testing whether rights-based narratives could affect discrimination without institutional backing: "Transgender people should have the same fundamental rights as others in India. They should have the right to housing, employment, and education without discrimination. All these rights that you have, they should also have."

3. Control video; $N = 1{,}132$. Participants received neutral information about voting rights: "Some people have the right to vote. If you have the right to vote, you can elect your representatives. That means you can choose who should be in power and who should make decisions on your behalf."

To maximize understanding, participants were given comprehension checks and corrected if needed.

E. Data Collection Phases and Samples

Data collection occurred in two phases (see fig. 3). Phase 1 (March–April 2023) focused on the main treatment effect, comparing private choices with three-person discussions. Phase 2 (May–July 2023) added the *no discussion (public)* and *two-person discussion* arms after receiving additional funding in order to understand the mechanisms behind the effects of the discussion.

When analyzing the data, I primarily use three different samples: (i) the *three-person discussion sample*, which measures the effect of the three-person discussion by including *no discussion (private)* and the *three-person discussion* arms in both phases 1 and 2; (ii) the *phase 2 sample*, which uses only phase 2 and includes all treatment arms to analyze the effect of the mechanism treatments; and (iii) the *video sample*, which includes all data from all phases since the videos were cross-randomized across all discussion arms.

F. Preanalysis Plan

I preregistered the design of both phases and document deviations from the preanalysis plan in appendix F. I changed the main specification to exclude interaction terms between discussion and video treatments to ease interpretation and increase power (I also show interacted specifications in table J7). The other main deviations were driven by unexpectedly low survey productivity in phase 1, which tightened budget constraints. These include (i) dropping a mixed-video arm in which each group

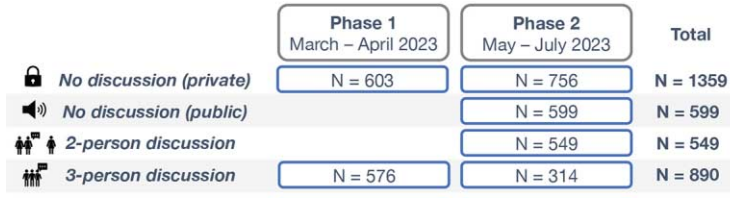


FIG. 3.—Sample sizes and timeline.

member saw different rights videos; (ii) delivering groceries after 2–9 weeks instead of 1 week; and (iii) collecting fewer mechanism outcomes.

G. Balance Checks

The treatment groups were well balanced on key characteristics (tables A4–A6), with joint *F*-tests showing no systematic differences from the control group for any treatment. As expected, given the large number of comparisons, individual variables show some statistically significant differences across treatment groups. *Three-person discussion* participants were more likely to have employed someone in the previous 2 years (table A4), and *rights messaging video* participants came from slightly larger households, with a slightly lower per capita food expenditure (table A5). I use LASSO to select all controls that predict either treatment status or outcomes (Belloni, Chernozhukov, and Hansen 2014; see app. sec. G.9), so these imbalances are unlikely to affect my results.

H. Outcome and Specification

The prespecified primary outcome is participants' individual choices in the outcome round of hiring. The design thereby estimates the causal effect of the discussion on participants' *postdiscussion* private discrimination. The outcome round occurred after group activities had finished, with 94% of respondents confirming that other group members could not hear their responses. While these choices were designed to be as private as possible, minimizing social image concerns, participants' choices may still have been affected by social image concerns if (i) they knew each other and would discuss choices later; (ii) they anticipated that neighbors would observe the delivery worker when the delivery took place; and (iii) enumerators observed the answers given by respondents. While I cannot rule out the first channel, for robustness I use an "extra private" outcome that addresses the second and third channels (see app. L).

The outcome round included six binary choices, two of which included a transgender worker. The main specification for participant i in group j, making a choice for the pair of workers k, is

$$\begin{aligned} ChooseAlternative_{ijk} = & \sum_{\tau \in \mathcal{T}} \beta_\tau \left(Treat_{\tau ij} \times Trans_{ijk} \right) + \gamma Trans_{ijk} + \sum_{\tau \in \mathcal{T}} \delta_\tau Treat_{\tau ij} \\ & + \mathbf{X}_{ijk} \boldsymbol{\Gamma}_0 + (\mathbf{X}_{ijk} \boldsymbol{\Gamma}_1 \times Trans_{ijk}) + \varepsilon_{ijk}. \end{aligned} \tag{1}$$

$ChooseAlternative_{ijk} = 1$ if i selects the *alternative* worker in pair k (who could be transgender or nontransgender) and is 0 when i selects the male *benchmark* worker. $Trans_{ijk} = 1$ if the alternative worker is transgender,

and is 0 if the alternative worker is nontransgender (cisgender male or female). The alternative worker is always compared with a male benchmark worker. $Treat_{\tau ij}$ is a dummy for whether i is in treatment arm $\tau \in \mathcal{T}$, where the set of treatments $\mathcal{T}$ is either (i) a dummy for the three-person discussion, (ii) dummies for each discussion-arm treatment, or (iii) dummies for each rights video.

Included in some specifications is $\boldsymbol{X}_{ijk}$ as a vector of controls. Controls are interacted with $Trans_{ijk}$ to control for differences in discrimination driven by observables. The controls include stratum fixed effects, differences in items offered, differences in reliability score, the benchmark worker's reliability score, whether the reliability score was shown, question order fixed effects, a dummy for whether the alternative worker was shown on the right, and data collection phase fixed effects. When analyzing the discussion-arm treatments, I control for the rights videos, and vice versa. I use double LASSO (Belloni, Chernozhukov, and Hansen 2014) to select additional controls that predict either the treatment or the outcome variables (see app. sec. G.9).

Throughout the paper, I define discrimination as the reduction in the probability that a worker is chosen because they are transgender (relative to nontransgender), conditional on other characteristics of the delivery options, such as the items on offer.

The main treatment effects are thus given by the coefficients β_τ, which describe the reduction in discrimination caused by the treatments. When interacted controls are not included, γ describes the control group's discrimination against transgender workers. Standard errors are clustered at the group-of-three level. For tables in the main text, I use randomization inference to calculate p-values (Young 2019). Since I have only one primary outcome, I do not correct it for multiple hypothesis testing.

I. Mechanism Outcomes

The experiment also measured several mechanisms (see app. G and the relevant results section for more detail on each).

Baseline measures.—Baseline measures included questions on (i) demographics; (ii) social desirability bias based on Crowne and Marlowe (1960) (phase 1 only); (iii) proximity of relationships between group members (phase 2 only); and (iv) a persuasiveness index to measure how persuasive an individual was likely to be in a discussion (phase 2 only). These questions were intermingled with questions about deliveries to reinforce the study's framing as market research.[16]

[16] I did not include baseline measures of anti-transgender attitudes or pretreatment hiring choices. While this would have increased power and yielded insights into the relationship between baseline attitudes and group discussions, it also risked undermining the

Treatment round choices.—Hiring choices during the *treatment round* are a prespecified secondary outcome. These allow me to examine what choices were made *during* (rather than *after*) the discussion.

Group observations.—During the group activities, an enumerator who was not facilitating the discussion noted observations about the group activity (e.g., who spoke first, who spoke in favor of a transgender worker). We recorded and transcribed the audio of the three-person discussions, and translated the Tamil transcriptions to English, yielding a dataset at the sentence × group × choice level.[17] The enumerator observations and audio transcripts are complementary since it was not possible to determine who said what in the transcripts.

Posthiring mechanisms.—Immediately after the hiring choices, we elicited further mechanism outcomes. Prespecified secondary outcomes were (i) predictions about the private hiring choices of other unknown people in the study; (ii) predictions about the private hiring choices of other participants in the same group; (iii) self-reported disapproval of discrimination when presented with discriminatory scenarios; (iv) a double list experiment (Droitcour et al. 2004; Glynn 2013) measuring the proportion of people agreeing with a discriminatory statement; and (v) questions about the legal status of transgender people. Additional exploratory mechanisms included (vi) beliefs about reliability (the probability that the worker will complete a delivery); (vii) hiring choices for a private grocery pickup unobservable to neighbors (phase 2 only); (viii) a measure of salience of the word "transgender" using a surprise recall task; (ix) two measures of participants' beliefs about the purpose of the study; and (x) self-reported reasons for their hiring choices (e.g., the most important factors when making their decision).

Follow-up survey.—When the delivery occurred, an average of 35 days after the initial survey (standard deviation [SD]: 14 days), we conducted a short (15-min.) survey to measure the persistence of treatment effects. As a prespecified secondary outcome, we asked six more hypothetical hiring choices with new worker photos and different grocery items. We made it clear to respondents that their choices would not result in actual deliveries.

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credibility of the main results for two reasons. First, asking additional questions about transgender people risked making the true purpose of the study more salient, exacerbating concerns of experimenter demand effects. This concern was especially severe for attitude questions that explicitly talk about discrimination against transgender people, contrasting with the hiring questions that are subtler and less obviously focused on discrimination. Second, evidence suggests that people prefer to act consistently with previous actions (Falk and Zimmermann 2017) and persuade themselves to ensure their preferences align with their previous actions (Schwardmann, Tripodi, and van der Weele 2022). If true, measuring baseline discrimination would anchor behavior to a pretreatment state and lead treatment effects to be underestimated.

[17] Consent for audio recordings was refused in 16% of discussions. The discussion's effects are not significantly different for groups that refused to be recorded (p of difference = .84).

As prespecified, I correct for multiple hypothesis testing within sets of secondary outcomes, namely, for attitudes (the list experiment and discrimination disapproval measure) and for norms (the predicted choices for community and own group).

IV. Results

A. *Effect of Three-Person Discussion*

The three-person discussion leads to large reductions in discrimination in the private choices made *after* the discussion in the later outcome round (table 1; fig. 4). In the control group, *no discussion (private)*, there is substantial discrimination: participants are 19 percentage points less likely to select a transgender worker than a nontransgender worker ($p < .001$; table 1, col. 1). But if participants were earlier involved in a group discussion and collective hiring decision, the probability that they chose a transgender candidate in their individual choices increases by 17 percentage points ($p < .001$). Participants in the discussion arm thus do not discriminate against transgender workers on average ($p = .30$).

The treatment effect of the three-person discussion is robust to the inclusion of controls (col. 2), to including only choices that involve a transgender worker (col. 3), and to restricting the sample to participants who did not see a video discussing transgender rights (table A7).[18]

To benchmark the size of the reduction in discrimination, I use the random variation in grocery items to infer the willingness to pay to avoid a transgender worker (fig. A9). In the *no discussion (private)* arm, participants are on average willing to sacrifice items worth Rs. 127 (approx. US$6.05 PPP) to avoid selecting a transgender worker, corresponding to 1.9 times the median daily per capita food expenditure in the sample. By contrast, in the *three-person discussion* arm, the willingness to pay to avoid a transgender worker is Rs. 13 (p of difference $< .001$) and is no longer significantly different from 0 ($p = .265$). The results therefore suggest that the discussion generates a large short-run reduction in discrimination.

The effect size is similar when examining only *costly* discrimination, that is, when participants avoid a transgender worker who offers more items, has a higher reliability score, or both (table A10, col. 1).[19] In the *no*

[18] Cross-randomized designs that do not account for interaction terms can yield incorrect inference (Muralidharan, Romero, and Wüthrich 2025). I show interaction terms for completeness (table J7), but since the results hold for participants who only saw the control video, interaction effects cannot be driving the main effects of the discussion.

[19] I do not find evidence of an implicit preference against transgender workers, in which they are discriminated against mostly when there is plausible deniability (e.g., when other attributes are different; de Quidt and Cunningham 2022). Transgender workers are not disproportionately favored when their attributes are the same as the male benchmark worker, compared with when they are dominated or dominating (table A10).

TABLE 1
EFFECT OF THREE-PERSON DISCUSSION ON PRIVATE CHOICES IN OUTCOME ROUND
(THREE-PERSON DISCUSSION SAMPLE, PHASES 1 AND 2)

	Chose Worker in Private Outcome Round (=1)		Chose Trans in Private Outcome Round (=1; pairs with trans only)
	(1)	(2)	(3)
Worker is trans × three-person discussion	.175***	.168***	
	(.022)	(2.2×10^{-2})	
	[<.001]	[<.001]	
Worker is trans	−.193***		
	(.013)		
	[<.001]		
Three-person discussion	−.004	.001	.167***
	(.011)	(1.0×10^{-2})	(.020)
	[.716]	[.912]	[<.001]
Relative number of items offered		.128***	.097***
		(5.0×10^{-3})	(.008)
		[<.001]	[<.001]
Relative reliability score		.017***	.013**
		(3.0×10^{-3})	(.005)
		[<.001]	[.012]
Reliability score is shown (=1)		.022***	.040***
		(8.0×10^{-3})	(.012)
		[.007]	[.001]
Number of observations	13,494	13,494	4,498
Number of participants	2,249	2,249	2,249
Number of groups	751	751	751
Outcome mean: no discussion (private), worker is nontrans[a]	.61	.61	
Outcome mean: no discussion (private), worker is trans[a]	.42	.42	.42
Controls		X	X
Controls interacted with worker is trans		X	

NOTE.—Standard errors are clustered at the group-of-three level and are in parentheses. Randomization inference *p*-values are in brackets. Unit of observation is the participant × choice level. Sample includes the *three-person discussion* arm and the *no discussion (private)* arm, in both phases 1 and 2. Column 3 includes only choices that involved a transgender worker. In cols. 1 and 2, the outcome is whether the *alternative worker* (vs. the male *benchmark worker*) was selected in the private choices in the *outcome round*. In col. 3, it is whether the transgender worker was selected. *Worker is trans* = 1 when the alternative worker is transgender and 0 when the alternative worker is male or female. The specification used is seen in eq. (1). Controls include stratum fixed effects; dummies for the rights videos; whether the individual was randomized into being offered three deliveries or one delivery, or was not part of this randomization; whether the alternative worker was shown on the right; phase fixed effects; and the controls selected by double LASSO (see sec. G.9 of the online appendix). In col. 2, controls are interacted with *worker is trans*, so the coefficient on *worker is trans* is not shown. Relative number of items offered is the number of items offered by the *alternative* worker minus the number of items offered by the male benchmark worker. Relative reliability score is the reliability score (out of 10) of the alternative worker minus the benchmark worker. *Reliability score is shown* is 1 when the reliability score is shown. Relative reliability score is coded as 0 when it is not shown.

[a] Dependent variable mean when worker is trans in *no discussion (private)* arm indicates transgender worker was selected (vs. male benchmark worker) 42% of the time. The mean when worker is male or female in the *no discussion (private)* arm is above 50% because participants on average prefer female alternative workers to male benchmark workers.

** $p < .05$.

*** $p < .01$.

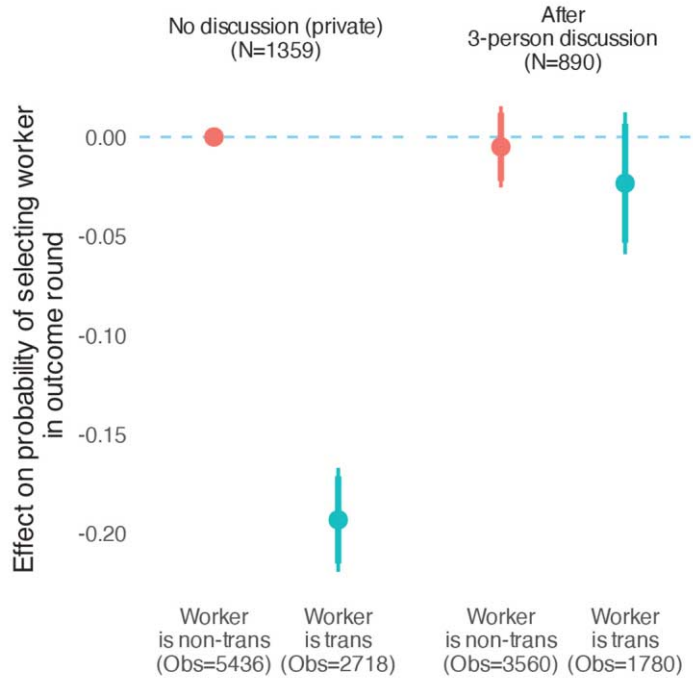


FIG. 4.—Effect of three-person discussion on private choices in outcome round. This graphically shows the main results from table 1. The *y*-axis shows the effect on the probability of choosing the *alternative* worker relative to the case where *worker is nontrans* in the *no discussion (private)* arm. The 90% and 95% confidence intervals are based on column 1 of table 1, based on standard errors clustered at the group-of-three level.

discussion (private) arm, even when shown a transgender worker who dominates on items or reliability score, or both, participants still select the nontransgender worker 47% of the time. By contrast, in the *three-person discussion* arm, this figure falls to 30% (difference: 17.5 percentage points, $p < .001$). The discussion also increases the probability of selecting a transgender worker when they offer fewer items or have a worse reliability score (12.7 percentage points, $p < .001$), implying that it also increases *positive* discrimination in favor of transgender workers.

There is heterogeneity in levels when separating the analysis of nontransgender workers into males and females (fig. A12).[20] Appendix I

[20] Female workers were the most preferred gender in both treatment conditions and were selected 72% of the time over the *benchmark choice* (who was always male). Male workers, always being compared with other males, were mechanically selected around 50% of the time. Transgender workers, however, were selected 42% of the time in the *no discussion (private)* arm but 59% in the *three-person discussion* arm. This implies that males were

shows further heterogeneity results, indicating that (i) participants trade off their preferences between items and worker; (ii) participants statistically discriminate against transgender workers, but the discussions' effects are not driven by this; and (iii) the discussion reduces discrimination by the same amount for male and female participants.

B. Effect of Transgender Rights Videos

The videos about transgender rights reduce discrimination by significantly less than the discussion (table 2). Both the rights messaging video and the legal rights video significantly reduced anti-transgender hiring discrimination, by 5.8 and 10.3 percentage points, respectively ($p < .01$). There is some evidence that the legal rights video has a stronger effect than the rights messaging video ($p \in [.01, .12]$, depending on the specification). Endorsing transgender rights thus appears to reduce discrimination somewhat more effectively when it is backed by the legal authority of the Supreme Court. This implies that the law can be an important tool for reducing societal discrimination and that raising awareness of the legal rights of minorities may be an underrated policy lever for addressing discrimination. However, the effect of top-down communication about the law is only 59% as large as the effect of the group discussion (p of difference $\in [.002, .04]$). Appendix J shows further results on the rights videos, including (i) a manipulation check showing that the legal rights videos affect people's beliefs about transgender rights; (ii) the interactions between the rights videos and the discussion; and (iii) how the rights videos affected the mechanism outcomes.

C. Persistence after 2–9 Weeks

To examine whether the effects of the discussion and the rights videos persist over the medium run, I ran a follow-up survey when the delivery was carried out. This survey took place an average of 35 days after the initial survey (SD: 14 days); 95.7% of the sample were found, and there was no evidence of differential attrition (table A15). For the follow-up, discrimination was measured using six hypothetical hiring choices designed to be as similar as possible to the main hiring choices. All of these questions were asked individually and in private. The questions used a new set of worker photos and new types of grocery items.

The three-person discussion led to reductions in discrimination in these hypothetical choices that were still present after 2–9 weeks (table 3). Participants were approximately 4 percentage points more likely to select

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preferred to transgender people in the control condition ($p < .001$), but transgender people were preferred to males in the treatment condition ($p < .001$).

TABLE 2
EFFECT OF RIGHTS VIDEOS ON PRIVATE CHOICES IN OUTCOME ROUND

	Chose Worker in Private Outcome Round (=1)		Chose Trans in Private Outcome Round (=1; pairs with trans only)
	(1)	(2)	(3)
Worker is trans	−.175***		
	(.016)		
	[<.001]		
Rights messaging video	−.013	−.016	.053***
	(.011)	(1.0×10^{-2})	(1.9×10^{-2})
	[.224]	[.103]	[.008]
Legal rights video	−.019*	−.021**	.081***
	(.011)	(1.0×10^{-2})	(1.9×10^{-2})
	[.078]	[.031]	[<.001]
Worker is trans × rights messaging video	.058***	.070***	
	(.023)	(2.1×10^{-2})	
	[.001]	[<.001]	
Worker is trans × legal rights video	.103***	.104***	
	(.022)	(2.0×10^{-2})	
	[<.001]	[<.001]	
Number of observations	20,382	20,382	6,794
Number of participants	3,397	3,397	3,397
Number of groups	1,134	1,134	1,134
Outcome mean: control video, worker is nontrans	.62	.62	
Outcome mean: control video, worker is trans	.45	.45	.45
Controls		X	X
Controls interacted with worker is trans		X	
p(rights messaging video = legal rights video)	.012	.055	.122
p(legal rights video = three-person discussion)	.024	.037	.002

NOTE.—Standard errors are clustered at the group-of-three level and are in parentheses. Randomization inference p-values are in brackets. Unit of observation is the participant × choice level. Sample includes all participants in both phases, in all discussion-arm treatments. Controls include dummies for the discussion-arm treatments. The specifications are otherwise the same as table 1.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

transgender workers in the hypothetical follow-up choices ($p = .03$). Approximately 25% of the short-run effect thus remained after around 1 month. This is comparable to persuasion decay rates seen in the political science literature (e.g., Hill et al. [2013], who estimate that 10%–15% of the initial effects of US TV political advertisements on voting remain after 30 days) although these effects are less persistent than those

TABLE 3
MEDIUM-RUN EFFECTS OF DISCUSSIONS ON HYPOTHETICAL HIRING CHOICES (2–9 WEEKS)

	Chose Worker in Follow-Up Round (=1)		Chose Trans in Follow-Up Round (=1; pairs with trans only)
	(1)	(2)	(3)
Worker is trans	−.127***		
	(.013)		
	[<.001]		
Three-person discussion	−.004	−.005	.043**
	(.011)	(.011)	(.019)
	[.776]	[.631]	[.026]
Worker is trans × three-person discussion	.054***	.048***	
	(.021)	(.021)	
	[<.001]	[.001]	
Number of observations	12,780	12,780	4,254
Number of participants	2,130	2,130	2,127
Number of groups	745	745	745
Outcome mean: no discussion (private), worker is nontrans[a]	.62	.62	
Outcome mean: no discussion (private), worker is trans	.49	.49	.49
Controls		X	X
Controls interacted with worker is trans		X	

NOTE.—Standard errors are clustered at the group-of-three level and are in parentheses. Randomization inference *p*-values are in brackets. Sample includes the *three-person discussion* arm and the *no discussion (private)* arm, in both phases 1 and 2. Controls include dummies for the rights videos, as well as the other controls specified in table 1. In the follow-up survey, workers in a pair always had the same reliability score and offered the same number of items. Specification is otherwise the same as table 1. As prespecified, for analysis, I drop the 0.4% of the sample who were randomly selected to actually receive a delivery from a transgender worker.

[a] In the no discussion (private) arm, transgender workers were selected over the male benchmark worker 49% of the time. The corresponding mean for nontransgender alternative workers exceeds 50% because participants on average prefer female alternative workers to male benchmark workers.

** $p < .05$.

*** $p < .01$.

reported by Broockman and Kalla (2016) and Kalla and Broockman (2020). By contrast, the videos about transgender rights did not lead to a detectable persistent effect on discrimination ($p \in [.12, .51]$; table A8).

Since the follow-up choices are hypothetical, the results are more vulnerable to concerns about experimenter demand effects and social desirability bias. However, experimenter demand effects are likely to be substantially *larger* for the rights videos (which increase the probability of successfully detecting the purpose of the experiment from 5% to 10%, $p < .001$; see app. L). The fact that there were no persistent impacts for rights videos thus suggests that the discussions' effects were not driven

by experimenter demand. The results therefore suggest that the large short-term effects of the discussion on discrimination may translate into medium-run effects. This raises the possibility that even short interventions involving horizontal within-group communication could have persistent effects on behavior.

V. Mechanisms: How Does the Discussion Affect Postdiscussion Discrimination?

How did group discussions reduce discrimination? I address this question in two sections. In this section, I use mechanism treatments and outcomes to identify how within-discussion behavior affected postdiscussion choices at all. I show experimental evidence in favor of interpersonal persuasion and suggestive correlational evidence that participants persuaded each other primarily by influencing perceived norms around hiring transgender workers, rather than by changing attitudes or beliefs. I then investigate why discussions created a large mean shift away from discrimination (sec. VI). Since standard models of communication (e.g., based on rational Bayesian agents) typically predict no changes in average beliefs, the substantial treatment effects require identifying asymmetries in the communication process. I provide suggestive evidence of three such asymmetries: pro-transgender participants spoke up more frequently, participants incorrectly updated and misperceived the strength of antidiscriminatory norms, and pro-trans arguments based on morality appear to have been particularly persuasive.

Appendix L also provides evidence against a number of alternative explanations of the results, including (i) other characteristics of the photos of workers, such as perceived caste; (ii) social image concerns that affect even private, postdiscussion choices; (iii) increased contemplation about choices; (iv) experimenter demand effects or social desirability bias; (v) increased salience of the notion of transgender; (vi) cheap talk due to low stakes; and (vii) discussion facilitators influencing discussions to be pro-trans.

A. *Candidate Mechanisms for Affecting Postdiscussion Behavior*

First, I explore three candidate mechanisms that could explain how behavior during the discussion could affect private postdiscussion discrimination.

1. *Norm-based persuasion.* Participants could persuade each other to discriminate less by changing perceived social norms. After observing frequent pro-trans statements and choices, participants may come to view discrimination as less socially acceptable or less prevalent. This perceived norm could then influence their postdiscussion

choices if they are motivated to adhere to social norms even in private.

2. *Attitude- or belief-based persuasion.*[21] Hearing pro-trans statements and choices could affect participants' personal attitudes or beliefs about transgender workers. For example, a participant could become more willing to interact with a transgender worker after a group member tells an anecdote of a positive interaction with a transgender person.
3. *Self-persuasion.* A participant may change her own attitudes toward transgender workers during the discussion, for example, because she selectively searches for arguments in favor of her original view and fails to fully adjust for this after the fact (Bénabou, Falk, and Tirole 2020; Schwardmann, Tripodi, and van der Weele 2022).

Below, I show experimental evidence in favor of interpersonal persuasion (consistent with point 1 or point 2). I also present correlational evidence that favors norm-based persuasion, while indicating that attitude- or belief-based persuasion is unlikely to be a key driver.

B. Evidence Consistent with Persuasion

I first present evidence that is compatible with both forms of interpersonal persuasion (norm- and attitude-based) but is not supportive of self-persuasion. Additional suggestive evidence is presented in appendix K.

Effect of listening to discussion.—Silent listeners in the *two-person discussion* treatment reduce discrimination as much as active discussants, suggesting that the effects are driven by what participants hear in the discussion from others (interpersonal persuasion), rather than through self-persuasion. Despite not speaking, listeners are 12.5 percentage points more likely to select a transgender worker after the discussion ($p < .001$; table 4, col. 2), an effect statistically indistinguishable from both two-person speakers ($p = .949$) and three-person discussion participants ($p = .377$). The effects persist in the 2–9 week follow-up ($\beta = 11$ percentage points, $p < .001$; table A20) and are similar for a short-run outcome that is completely private, that is, unobserved by neighbors and enumerators ($\beta = 11$ percentage points, $p = .01$; table L10; see app. L). These results suggest that the effects are maintained even when postdiscussion behavior is private, consistent with persuasion mechanisms.

[21] I denote both norm- and attitude-based influence as persuasion, since both mechanisms make very similar behavioral predictions: they both involve participants saying and choosing things that influence others to discriminate less *in private* after the discussion ends. This follows previous literature that defines persuasion as when a sender sends a message to a receiver with a potential interest in affecting the receiver's behavior (e.g., DellaVigna and Gentzkow 2010) and work that includes *social proof* as a key mechanism for persuasion (Cialdini 2009).

TABLE 4
EFFECT OF MECHANISM TREATMENTS (LISTENER, PUBLIC, OBSERVER)

	Chose Trans (=1; pairs with trans only)	
	Treatment Round (1)	Outcome Round (2)
No discussion (public), pooled	.015	
	(.022)	
	[.379]	
Observer (no discussion, public)		.043**
		(.026)
		[.041]
Nonobserver (no discussion, public)		.022
		(.031)
		[.190]
Speaker (two-person discussion)	.189***	.131***
	(.030)	(.028)
	[<.001]	[<.001]
Listener (two-person discussion)		.125***
		(.033)
		[<.001]
Three-person discussion	.229***	.176***
	(.033)	(.031)
	[<.001]	[<.001]
Number of observations	4,066	4,436
Number of participants	2,033	2,218
Number of groups	740	741
Outcome mean: no discussion (private), worker is trans	.49	.41
p(public = speaker)	<.001	
p(public = three-person discussion)	<.001	
p(speaker = three-person discussion)	.534	.118
p(observer = nonobserver)		.506
p(observer = speaker)		.002
p(observer = listener)		.029
p(observer = three-person discussion)		<.001
p(speaker = listener)		.949
p(listener = three-person discussion)		.377

NOTE.—Standard errors are clustered at the group-of-three level and are in parentheses. Randomization inference p-values are in brackets. Sample includes all treatment arms in phase 2 of data collection. In col. 1, the *listeners* are excluded (since they only make choices after they have observed the discussion). The specification used is otherwise the same as in tables 1 and 2, cols. 3. *No discussion (public), pooled* includes both *observers* and *nonobservers*, since both these types are treated the same until after the treatment round.

** $p < .05$.

*** $p < .01$.

Heterogeneity by pro-trans discussion.—Participants who hear more pro-transgender statements from others discriminate less afterward: each time a participant hears a group member speak positively about a transgender option (out of a maximum of four, according to enumerator observations) is associated with a 6.2 percentage point reduction in discrimination

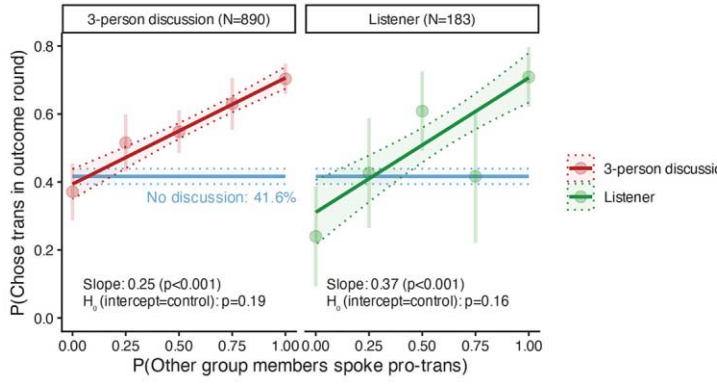


FIG. 5.—More pro-trans discussions have more positive effects on postdiscussion discrimination. *P(Other group members spoke pro-trans)* (*x*-axis) is the proportion of (discussion choice × participants) that spoke positively about a transgender option. There were two other group members, each with two opportunities to speak positively about a transgender worker. *P(Chose trans in outcome round)* (*y*-axis) is the probability a participant chose the transgender worker in the private outcome round. Plotted line of best fit and 95% confidence intervals are based on raw data with no controls. *A*, Three-person discussion participants in phases 1 and 2; *B*, Listeners in phase 2. Slope and *p*-values come from models with LASSO controls and all other controls in table 1. $\beta = 0.250$ implies that one additional favorable participant–choice opportunity out of four corresponds to an approximately 6.25 percentage point increase in the probability of selecting a transgender worker in the outcome round ($100 \times 0.250/4 = 6.25$). H_0(intercept = control) tests whether a discussion participant for whom *P(Other group members spoke pro-trans) = 0* exhibits different postdiscussion discrimination from the control group.

after the discussion ($p < .001$; fig. 5).[22] This pattern also holds for silent listeners (9.3 percentage points, $p < .001$; right panel), whose results are more likely to reflect causal effects since they have minimal opportunity to endogenously affect their group members' behavior. These correlational results are consistent with either norm-based persuasion (since more pro-trans discussions more strongly signal a pro-trans norm) or attitude or belief-based persuasion (since more pro-trans arguments are more likely to change attitudes or beliefs).

C. *Evidence for Norm-Based Persuasion*

Having shown that the results are driven by participants influencing each other's postdiscussion behavior, I now document descriptive evidence in favor of norm updating as the primary channel.

Effect of discussion on within-group norms.—Discussions had large effects on participants' beliefs about their peers' behavior, providing strong evidence for norm updating. In an incentivized prediction task, participants

[22] The coefficients on pro-trans discussions are not significantly different when also controlling for the probability of hearing anti-trans discussions.

predicted whether their two group members selected transgender workers in private postdiscussion choices (see app. G). The discussion increased these predicted probabilities by 24 percentage points (table I3, panel A, col. 2; $p < .001$). This change in perceived within-group norms appears to be a key mediator of the main treatment effects: controlling for within-group norms significantly weakens the effect on private discrimination (table A19), while controlling for other mechanism outcomes (community-wide norms, disapproval of discrimination, beliefs about reliability) does not. Section VI will discuss explanations for why norms shift so starkly away from discrimination.

Effect of observing others' choices.—To isolate whether norm updating alone is sufficient to reduce discrimination, I examine the observers in the *no discussion (public)* treatment. Simply observing others' public choices (without a discussion) increases predictions that group members will select transgender workers in private by 4.6 percentage points ($p = .08$; table 5, col. 1), correcting participants' overestimation of discrimination (p-value of misperception = 0 is .17).[23] The estimated reduction in discrimination in the private round is similar in size ($\beta = 4.2$ percentage points, $p = .07$; col. 3).[24] In other words, simply observing others' choices without a discussion leads to small updates in perceived within-group norms, and this is sufficient to reduce later private discrimination. While this demonstrates that norm updating can reduce discrimination, the discussions' effects are larger. This plausibly reflects the sources of asymmetry documented in section VI: for example, pro-trans participants' greater willingness to speak up likely leads to larger shifts toward an anti-discriminatory norm.

D. Suggestive Evidence against Attitude- or Belief-Based Persuasion

The available descriptive evidence does not favor attitude- or belief-based persuasion.

Attitudes.—Discussions produce only small or null effects on measures of personal attitudes toward transgender workers, indicating that effects are more likely to be operating through norm updating than through changes in attitudes. The proportion of participants who said that (in their opinion) discrimination was wrong in response to two vignettes increased only slightly from 92.8% to 94.4% (p of difference: .05; table I3, panel B,

[23] There is no statistically significant average effect of making choices public on the probability of selecting a transgender worker in the treatment round ($\beta = 1.5$ percentage points, $p = .38$; table 4, col. 1), so observing public choices in the no discussion (public) arm is on average equivalent to observing the control group's choices.

[24] The observer–control difference in the association between peers' treatment-round choices and the participant's private outcome-round choice is 6.8 percentage points per additional favorable peer choice, out of four peer-choice opportunities ($100 \times 0.272/4 = 6.8$; $p < .001$; col. 4).

TABLE 5
EFFECT OF OBSERVING OTHERS' CHOICES

	Predicted Probability of Others Selecting Trans (within-group, outcome round)		Chose Trans in Private Outcome Round (=1)	
	(1)	(2)	(3)	(4)
Observer (no discussion, public)	.046*	−.179***	.042*	−.098**
	(.026)	(.045)	(.023)	(.043)
	[.079]	[<.001]	[.072]	[.024]
π_{-i} = p(others in group chose trans in treatment round)		.056		−.040
		(.050)		(.051)
		[.263]		[.431]
π_{-i} × observer (no discussion, public)		.440***		.272***
		(.085)		(.079)
		[<.001]		[<.001]
Number of observations	2,289	2,289	2,310	2,310
Number of participants	1,148	1,148	1,155	1,155
Number of groups	452	452	453	453
Outcome mean: no discussion (private)	.32	.32	.41	.41
Controls	X	X	X	X

NOTE.—Standard p-values are in brackets. Unit of observation is the participant × choice level. Outcome in cols. 1 and 2 is the predicted probability that other group members select a transgender worker in private. Outcome in cols. 3 and 4 is whether participant selected a transgender worker themselves in the outcome round. In all columns, the sample is only the *observers (no discussion, public)* and the control group (*no discussion, private*) in phase 2. *P(others in group chose trans in treatment round)* (π_{-i}) is the proportion of times (out of a maximum of four) that the other two participants in the group selected a transgender worker in the treatment round. All regressions control for p(chose trans in treatment round), the proportion of times (out of a maximum of two) that the participant herself selected a transgender worker in the treatment round. Other controls include stratum fixed effects; dummies for the rights videos; whether the alternative worker was shown on the right; relative number of items offered by the transgender worker; relative reliability score; and a dummy for whether the reliability score was shown.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

col. 2). Controlling for this measure of private attitudes does not weaken the effects of discussion on discrimination ($p = .92$; table A19). There are also no effects on a double list experiment (Droitcour et al. 2004; Glynn 2013) that measured how many people agreed with the statement, "In general, if I see a transgender person, I walk away" ($p = .22$; table I3, panel B, col. 1).

Beliefs about reliability.—Belief-based (statistical) discrimination appears to underlie some of participants' unwillingness to select transgender people, driven by negative stereotypes that portray transgender workers as unreliable. However, the effect of the discussion does not appear to be driven by changes in such statistical discrimination. The discussion does not significantly affect beliefs about the reliability of transgender workers ($p = .15$; table I3, panel A, col. 3); controlling for these beliefs does not

weaken the effect on discrimination (table A19); and I find no evidence that the three-person discussion reduces the belief-based component of discrimination, although I am underpowered for this latter test (table I4, col. 2).

VI. Mechanisms: Sources of Asymmetry

Having shown evidence in favor of norm-based persuasion as a key channel for the effects on postdiscussion behavior, I now examine why discussions produce a large average shift away from discrimination.

Standard models of communication in economics predict that communication should lead behavior to converge and (absent misperceptions in participants' priors) should not change average behavior.[25] The substantial reduction in discrimination therefore requires asymmetries in the discussion process. I identify three key asymmetries: pro-transgender participants are disproportionately vocal, participants fail to correctly account for this selective communication when inferring norms, and moral arguments against discrimination appear to be more persuasive than anti-trans arguments. Much of the evidence on these asymmetries comes from heterogeneity analyses rather than exogenous mechanism treatments; these results should be interpreted with some caution.

A. *Behavior during Discussion and Willingness to Speak Up*

The first asymmetry I document is that pro-transgender participants are more vocal than others, leading to discussions that were strongly in favor of transgender workers. Their stronger advocacy appears to be partly driven by a stronger desire to influence other group members' choices.

Choices during the discussion (treatment round).—The choices made *during* the discussion are favorable toward transgender workers. Participants were 20 percentage points more likely to select a transgender worker in the collective choices during the discussion than in the private choices made by *no discussion (private)* participants (table A16; $p < .001$). In the discussion choices, there was even positive discrimination in favor of transgender workers relative to nontransgender workers (11 percentage points; $p < .001$).

Pro- and anti-trans statements in discussion.—Participants also on average communicated about transgender workers in a positive way. Based on

[25] Suppose each participant i is learning about their own (dis)utility θ_i from hiring a transgender worker. Under a standard Bayesian communication model (with a common prior and truthful information exchange), participants converge to a common posterior (Aumann 1976; Geanakoplos and Polemarchakis 1982). The martingale property of Bayesian updating also implies that the posterior belief (and therefore postcommunication behavior) will not shift on average across participants relative to the prior *if* the prior is correctly specified, i.e., it accurately represents the true data-generating process.

transcripts with a sentence-level indicator of arguments in favor of or against the transgender worker, participants were 2.5 times more likely to make pro-trans than anti-trans statements (38% of sentences were pro-trans vs. 15% anti-trans; p of difference: <.001).

Substantial minority of pro-trans participants.—A substantial minority of participants favor transgender workers despite strong average discrimination. Figure A17 presents individual-level estimates of willingness to pay (WTP) for transgender workers in the outcome round, derived from a regression with individual fixed effects. The distribution reveals wide variance, so despite a substantial negative WTP on average, 31% of control participants have positive WTP for transgender workers. This sizable pro-transgender faction is likely to be important for explaining the pro-transgender nature of group discussions.

Pro-trans participants are more vocal.—This pro-trans pattern in the discussion appears to be partly explained by pro-trans individuals choosing to be more vocal. I use private choices after the discussion as a proxy for baseline pro-trans behavior and show that this is correlated with dominating the discussion of transgender workers (table A21).[26] For each additional transgender worker a participant selected in the private round, they were 11 percentage points more likely to speak first (32%, $p = .03$) and 15 percentage points more likely to dominate discussions (27%, $p = .02$) when discussing transgender workers (according to enumerator observations). No such correlation exists for nontransgender worker discussions, implying that pro-trans participants selectively spoke up about transgender workers, despite not being more talkative in general. This asymmetry could plausibly be a driver of reductions in postdiscussion discrimination.

1. Why Do Pro-Trans Participants Speak Up More?

Pro-trans participants might have spoken up more because of *social image concerns* (e.g., anti-trans participants self-censor because they did not want to appear discriminatory) or because they had greater *influence concerns* (e.g., pro-trans participants cared more about stopping others taking discriminatory actions; for a framework describing these motives, see Bénabou, Falk, and Tirole 2020). Suggestive evidence from mechanism outcomes indicates that influence concerns were likely more important in driving pro-trans participants' advocacy.

[26] Since I purposely did not collect baseline measures of discrimination (to minimize priming and experimenter demand effects), the evidence here should be taken as suggestive. However, under a reasonable monotonicity assumption, the postdiscussion discrimination will correlate positively with baseline discrimination: as long as participants are not too effective at persuading each other, initially anti-trans participants are unlikely to become more pro-trans than those who were pro-trans from the start.

Social image concerns.—There is no strong evidence for social image concerns driving pro-trans participants' disproportionate willingness to speak up. First, the *no discussion (public)* treatment did not reduce discrimination in the treatment round (table 4, col. 1; $p = .38$), implying that exogenously increasing social image concerns by revealing choices to other group members did not reduce discrimination.[27] This suggests that social image concerns are unlikely to be a key driver of pro-trans communication in the discussions, although importantly I cannot rule out that social image concerns have a stronger effect in discussions than when silently making choices (e.g., because the social costs of speaking are larger than choosing or because of dynamically increasing social image concerns after pro-trans statements). Second, pro-trans advocates were not more likely to report making statements "that others would agree with" or in order "to look good" in postdiscussion debriefs compared with those who did not advocate for transgender workers (fig. A22, bottom panels; $p = .76$ and .35).

Influence concerns.—The evidence instead hints at an aim to persuade others to change their behavior. Participants who spoke in favor of transgender workers were 8.4 percentage points more likely to report trying to get others to agree with them (13%, $p = .03$; fig. A22, top left). While these differences are only correlational and do not extend to all measures of persuasion (fig. A22, top right), they suggest pro-trans participants disproportionately engaged in the discussion to shape others' choices. Pro-trans participants' behavior may be more driven by persuasion concerns than anti-trans participants' behavior for a number of reasons: they may have a more intense preference in favor of transgender workers, or they may care more about others' behavior (e.g., if they perceive discrimination as a social wrong, rather than a personal preference). Alternatively, pro-trans participants may recognize their rhetorical advantage (Schkade, Sunstein, and Kahneman 2002): as I show below (sec. VI.C), they deploy moral arguments that may be especially compelling, thus making their active participation in the discussion more attractive.

B. Incorrect Inference about the Norm

The second source of asymmetry that could explain the large shift toward an antidiscriminatory norm is that people did not make correct inferences, tending to *over-update* on pro-trans behavior and to *under-update* on anti-trans behavior during the discussion. For example, if participants did not account for pro-trans speakers' disproportionate willingness to speak up, the discussion would make them overestimate their group's

[27] The null effect is not driven by participant inattentiveness, because participants' choices in *no discussion (public)* did react: they converged within groups, despite no mean effects (p-value of effect on intracluster correlation within groups: .10).

true support for transgender workers. This mechanism builds on theories of misperceived norms (Bursztyn, Egorov, and Fiorin 2020). But here the norm is misperceived because of the conversation itself. I find two main pieces of suggestive evidence of such incorrect inference.

Silence neglect.—Participants appear to have neglected the informational value of silence during discussions, accentuating updates toward pro-transgender norms. While participants strongly updated their beliefs after hearing pro-transgender statements, they failed to interpret silence as evidence of anti-transgender attitudes. When no group members made pro-transgender statements, participants' predictions about their peers' choices remained indistinguishable from control group predictions (p of difference: .22; fig. A11). This implies that participants did not interpret silence as a signal of anti-trans attitudes, even though silence often indicated opposition rather than indifference to selecting transgender workers. The failure to interpret silence as informative carried through to postdiscussion behavior: three-person discussion participants who heard no pro-trans statements did not increase discrimination relative to the control group (fig. 5).[28] This evidence is in line with theories of the "spiral of silence," in which people come to believe that a view is more prevalent than it truly is because of selective communication (Noelle-Neumann 1974; Ho and Huang 2026), and work on selection neglect, showing that people fail to discount how information is filtered before reaching them (Enke 2020).[29]

Overcorrection of misperception.—The evidence is consistent with the discussion's effects partly stemming from correcting—and then overcorrecting—initially misperceived norms. I do find evidence of an initial misperception: the control group underestimated how often their peers would privately choose transgender workers (by 5.3 percentage points, $p < .001$; fig. A18). But discussions did more than fix this misperception—they created a new one in the opposite direction. After discussions, participants overestimated their peers' selection of transgender workers by 3.0 percentage points ($p = .04$). This pattern is also consistent with a spiral of silence mechanism: pro-trans participants speak up more, other participants do not sufficiently account for this, and thus those other participants overestimate support for transgender workers among their peers.

[28] For listeners, the raw data suggest that discussions with no pro-trans statements may have slightly decreased the predicted and actual probability of choosing a trans worker (figs. 5, A11, right-hand panels). The intercept is not significantly different from 0 in either case when controls are included ($p = .16$ and .22), but the coefficient is negative. While the sample size of listeners is small, this may indicate *partial* rather than total silence neglect.

[29] Unlike the standard spiral-of-silence mechanism (that emphasizes social image concerns), participants' desire to persuade others appears to be the primary driver of selective communication in this context.

C. *Rhetorical Asymmetry and Moral Arguments*

The discussion transcripts reveal a third source of asymmetry: the types of arguments participants used and how persuasive they appear to be. Protrans advocates more frequently invoked moral concepts (e.g., equality, rights, and opportunity) when discussing transgender workers, and this moral framing is strongly associated with lower postdiscussion discrimination. While this pattern is only correlational, it suggests a form of *rhetorical asymmetry* (Schkade, Sunstein, and Kahneman 2002) in which protransgender arguments were more persuasive than anti-transgender ones. The pattern emerges consistently across multiple analyses: a sentence-level transcript analysis, AI-generated hypotheses about discussion characteristics, and enumerator observations of discussions. The relationship holds for silent listeners, for which causality is more plausible, and even after controlling for within-discussion choices. These findings align with psychological research on moral reframing, which shows that arguments framed in moral terms are particularly persuasive when they align with listeners' values (Voelkel and Feinberg 2018; Feinberg and Willer 2019). Combined with the evidence on norm updating, this suggests that moral arguments may serve as powerful signals of antidiscriminatory norms.

Transcript sentences.—Sentence-level semantic analysis of the three-person discussions suggests that moral language predicts reductions in postdiscussion discrimination. Using OpenAI's text-embedding-3-small model, I use *k*-means clustering to classify sentences based on similarity into 20 groups (table A13; see app. sec. H.1 for methodology). Sentences about "giving opportunities" to a worker and sentences that explicitly acknowledge that a worker is transgender are especially common for choices involving a transgender worker.[30] These sentences also strongly predict selecting transgender workers privately after the discussion: a 1 SD increase in "giving opportunities" sentences corresponds to a 6.8 percentage point decrease in postdiscussion discrimination ($q = 0.001$, cluster 1), and a 1 SD increase in explicitly acknowledging transgender identity predicts a 5.7 percentage point increase in the probability of selecting a transgender worker ($q = 0.001$, cluster 2). In contrast, discussions that focused on workers' appearance predict weaker treatment effects ($\beta = -7.3$ percentage points, $q = 0.003$, cluster 3), in line with stereotypes portraying transgender people as looking strange or untrustworthy.

Broader transcript features.—To analyze higher-level characteristics of the discussion that are correlated with reduced discrimination, I follow

[30] Indeed, the transcript quotes that were *most* uniquely representative of transgender discussions (relative to nontransgender discussions, based on relative cosine similarity of embeddings) were "They are transgender, therefore, they deserve an opportunity" and "We should give them equal rights too." Participants were especially likely to use words like "opportunity," "chance," and "rights" (see word cloud in fig. A24).

recent work that uses AI to generate hypotheses (Batista and Ross 2024; Ludwig and Mullainathan 2024). Specifically, I (i) use a large language model (LLM) to generate 500 hypotheses about the main difference between two randomly selected transcripts (without revealing any postdiscussion outcomes); (ii) use an LLM to rate all transcripts based on how much they satisfy each hypothesis; (iii) combine hypotheses whose ratings are highly correlated using factor analysis; and (iv) test whether the aggregate rating is correlated with postdiscussion discrimination (see app. sec. H.2).

Using this method, I find that discussions that explicitly acknowledge the transgender worker's identity and place a stronger emphasis on social equity and support for marginalized individuals are strongly associated with less discrimination ($q < 0.001$; fig. A14, rows 1 and 2). A focus on gender identity and on grocery items during the transgender discussion is also associated with less postdiscussion discrimination ($q < 0.05$, rows 3 and 4). In contrast, discussions about transgender workers that focus on other details about the worker (e.g., the language proficiency or delivery performance) are associated with weaker treatment effects ($q = 0.07$ and 0.03), in line with the view that moral arguments are more persuasive than practical ones in this context. Notably, there do not seem to be moral anti-transgender arguments that are persuasive: anti-trans participants only resort to details about the workers' qualifications.

Enumerator observations.—Enumerators' observations are consistent with this picture. When discussing transgender workers, participants more frequently cite moral rationales (e.g., wanting to give an opportunity to or help the worker; saying that the worker is "also human") compared with discussions of nontransgender workers (difference = 32.5 percentage points, $p < .001$; fig. A23). And these moral rationales correlate with lower discrimination after the discussion: listeners who heard them were 6 percentage points more likely to select transgender workers afterward ($p < .001$).[31]

Moral discussions.—The evidence above suggests that discussions focusing on giving opportunities, social equity, or other moral notions are associated with reductions in postdiscussion discrimination, in keeping with the psychological literature on moral reframing (Voelkel and Feinberg 2018; Feinberg and Willer 2019). To more directly test the hypothesis that moral reframing reduces discrimination, I use an LLM to rate how much each discussion transcript invoked morality on a 0–1 scale

[31] The increase in pro-social reasoning translates to participants' reported reasoning in the private outcome round. When participants were asked why they made their outcome-round choices, those who had been involved in a discussion were 11 percentage points (24%) more likely than the control group to cite pro-social reasons for their choices ($p = .01$).

(see app. sec. H.3). Participants used moral language substantially more when discussing transgender workers (from 0.18 to 0.44, +0.99 SD, $p < .001$). The association is 39 percentage points in column 1 and remains positive after controlling for pro-trans statements or treatment-round choices (table 6). While this evidence is only correlational, it suggests that the moral content of discussions may reduce discrimination independently of the within-discussion choices. Moral language moreover appears to operate through the *norm updating* channel described above, that is, by affecting participants' perceptions of the norm of behavior toward transgender workers: a 1 SD increase in moral discussion is also associated with large updates in within-group norms (14 percentage points when controlling for treatment round choices, $p = .02$).

TABLE 6
MORALITY RATING OF DISCUSSION PREDICTS REDUCED POSTDISCUSSION DISCRIMINATION

	Chose Trans in Outcome Round (=1)		
	(1)	(2)	(3)
Three-person discussion	−.008	−.094***	.027
	(3.3×10^{-2})	(3.5×10^{-2})	(2.7×10^{-2})
	[.799]	[.008]	[.327]
Discussion morality score, trans choices (Z)	.394***	.287***	.116**
	(5.4×10^{-2})	(5.8×10^{-2})	(4.8×10^{-2})
	[<.001]	[<.001]	[.016]
Discussion morality score, nontrans choices (Z)	−.016	−.007	.024
	(7.3×10^{-2})	(7.2×10^{-2})	(6.8×10^{-2})
	[.831]	[.924]	[.728]
p(others in group spoke pro-trans in discussion)		.215***	
		(4.3×10^{-2})	
		[<.001]	
Number of times group chose trans in treatment round			.214***
			(1.1×10^{-2})
			[<.001]
Number of observations	4,362	4,362	4,362
Number of participants	2,181	2,181	2,181
Number of groups	728	728	728
Outcome mean: no discussion (private)	.42	.42	.42

NOTE.—Sample includes "three-person discussion" and "no discussion (private)" participants, so the control group is the omitted category. "Discussion morality score" is the morality score of the discussion transcripts rated by an LLM, designed to measure how much morality is invoked (see sec. VI.C; app. H). Each discussion-choice transcript receives a rating. I average these ratings for each group's transgender and nontransgender choices separately, normalizing to ensure a mean of 0 and standard deviation of 1. "Others in group: proportion of discussions spoke in favor of trans" is the proportion of times (out of a maximum of four) the other group members spoke positively about a transgender option. "Number of times group chose trans in treatment round" is the number of times a transgender worker was chosen by the group in the treatment round. All four of these additional predictors are coded as 0 for the control group.

** $p < .05$.

*** $p < .01$.

Heterogeneity on persuasiveness.—One piece of suggestive evidence against persuasion and rhetorical asymmetry is that participants who rate their group members as more persuasive in a prediscussion question do not discriminate less afterward. Neither listeners nor discussion participants discriminate less when their peers score higher on a prediscussion persuasiveness index ($p = .61$ and .91; table A25). This measure has limitations—it captures perceptions of a persuasive personality before the discussion, rather than transgender-specific advocacy or likelihood of speaking up. The null result thus does not preclude effects stemming from pro-trans participants being more willing to speak up or from selecting more persuasive moral arguments specifically when discussing transgender workers. Nevertheless, the null result suggests caution in attributing effects entirely to rhetorical asymmetry.

VII. Discussion

A group discussion about hiring transgender workers sharply reduced discrimination in subsequent real-stakes hiring choices. Despite lasting only 10 minutes, the discussion also decreased discriminatory choices several weeks later. I show experimental evidence that the impact is driven by interpersonal persuasion and suggestive evidence based on correlations that it stems from pro-transgender participants speaking up more frequently during discussions, leading others to perceive social norms as less discriminatory and consequently reduce their own private discrimination after the discussion ended. These results provide a proof of concept that horizontal communication can lead to large reductions in discrimination. Progressive social change can thus occur not only by generating social contact between in-groups and out-groups but also by encouraging communication among the in-group in the right circumstances.

Why did horizontal communication sharply reduce discrimination in this setting? It can also amplify prejudice, as in 1930s Germany when anti-Semitic sentiment spread through social networks (Satyanath, Voigtländer, and Voth 2017) or in other psychological studies examining group dynamics (Myers and Bishop 1970). The norm-based persuasion mechanism I show evidence for is directionally neutral—it could either reduce or increase discrimination depending on which voices dominate the discussions. The central puzzle is thus explaining why pro-trans participants were so vocal in this context.

The contextual factor that appears to be key is a striking disconnect: while discrimination is strong and common in the absence of communication, there is nevertheless widespread agreement that discrimination is wrong (e.g., 93% of control participants say it is wrong). This contrasts with many settings where discrimination is not only common but also widely accepted or even morally encouraged. This gap between behavior

(descriptive norms) and moral beliefs (prescriptive norms) could explain four key patterns that could drive the positive effects: (i) the substantial minority of pro-trans participants; (ii) the minimal social costs for advocating against discrimination, unlike in contexts where discrimination is both practiced and accepted; (iii) pro-trans participants' greater propensity to try to influence others, consistent with them viewing discrimination as a moral wrong rather than a personal choice and thus caring more about others' actions; and (iv) the particular prevalence and influence of pro-trans moral arguments about equal rights and fairness that align with participants' stated values (unlike in Nazi Germany, where moral narratives focused on impurity were used to *encourage* discrimination). All of these factors would, in turn, contribute to pro-trans participants' disproportionate willingness to speak up and thus plausibly explain the reductions in postdiscussion discrimination. They could also explain why the descriptive norm seems to be so responsive to the discussion. Future research could valuably explore the boundary conditions of when horizontal communication is helpful in other contexts. Looking for other contexts with a wedge between the level of discrimination (what people do) and the acceptance of discrimination (what people say is right), for example, using attitude surveys, would be a promising avenue for finding such contexts.

The study has several important limitations. First, the main effects are measured only shortly after the discussions, which may explain the large coefficients, and the medium-run effects are based only on hypothetical choices. The postdiscussion decisions are also of the same type as those made during the discussions. This raises the concern that some of the effects may be driven by participants' drive to maintain consistency (although the large effects on silent listeners alleviate this concern, since they did not make choices in the discussion). Second, I do not elicit baseline measures of discrimination. While this was done deliberately to minimize the risk that participants guessed the purpose of the experiment and reduce experimenter demand effects, it limits the ability to conclusively understand which types of participants speak up and how group composition relates to persuasion. Third, the communication involved in the discussion is not entirely natural: the presence of an enumerator facilitating the discussion may influence participants' behavior, and participants make incentivized choices at the same time as communicating. Although facilitator influence may also exist for the control group, and the intervention resembles a policy-style intervention, the results may not as clearly map onto the effects of organic horizontal communication in society absent an intervention. Fourth, while the evidence in favor of interpersonal persuasion is based on an experimental variation, the evidence on other mechanisms related to norm updating and asymmetries is only suggestive and primarily based on consistent patterns of correlations. Future research could explore

participants' decisions to speak up in more organic scenarios (e.g., with no facilitator) and use exogenous variation in group composition and baseline attitude measures to better understand mechanisms.

Future research could also explore how to design policies based on the insight that in-group communication can reduce discrimination under the right circumstances. Policies that create discussions at scale to change attitudes are particularly promising (e.g., Broockman and Kalla 2016; Kalla and Broockman 2020; Dhar, Jain, and Jayachandran 2022). The role of vocal pro-trans participants also suggests it may be possible to amplify the voice of pro-social actors within a network by identifying them and training them to influence others' behavior. This approach is used by activist nongovernmental organizations and has shown promise when trying to reduce the stigma surrounding menstruation (Macours, Rueda, and Webb 2024). Finally, we could use groups rather than individuals to make high-stakes decisions in order to reduce discrimination in cases where a pro-minority prescriptive norm is in place.

Data Availability

Code replicating the tables and figures in this article can be found in Webb (2026) in the Harvard Dataverse, https://doi.org/10.7910/DVN/UC27B9.

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